Lord and Lady of the Cemetery (late 19th–early 20th century) in Nechung Chapel. Photograph: Thomas Laird

Has examples of

Local adaptation

Divergence

Homeostatic & acclimatization responses

Maladaptation

Human evolutionary history

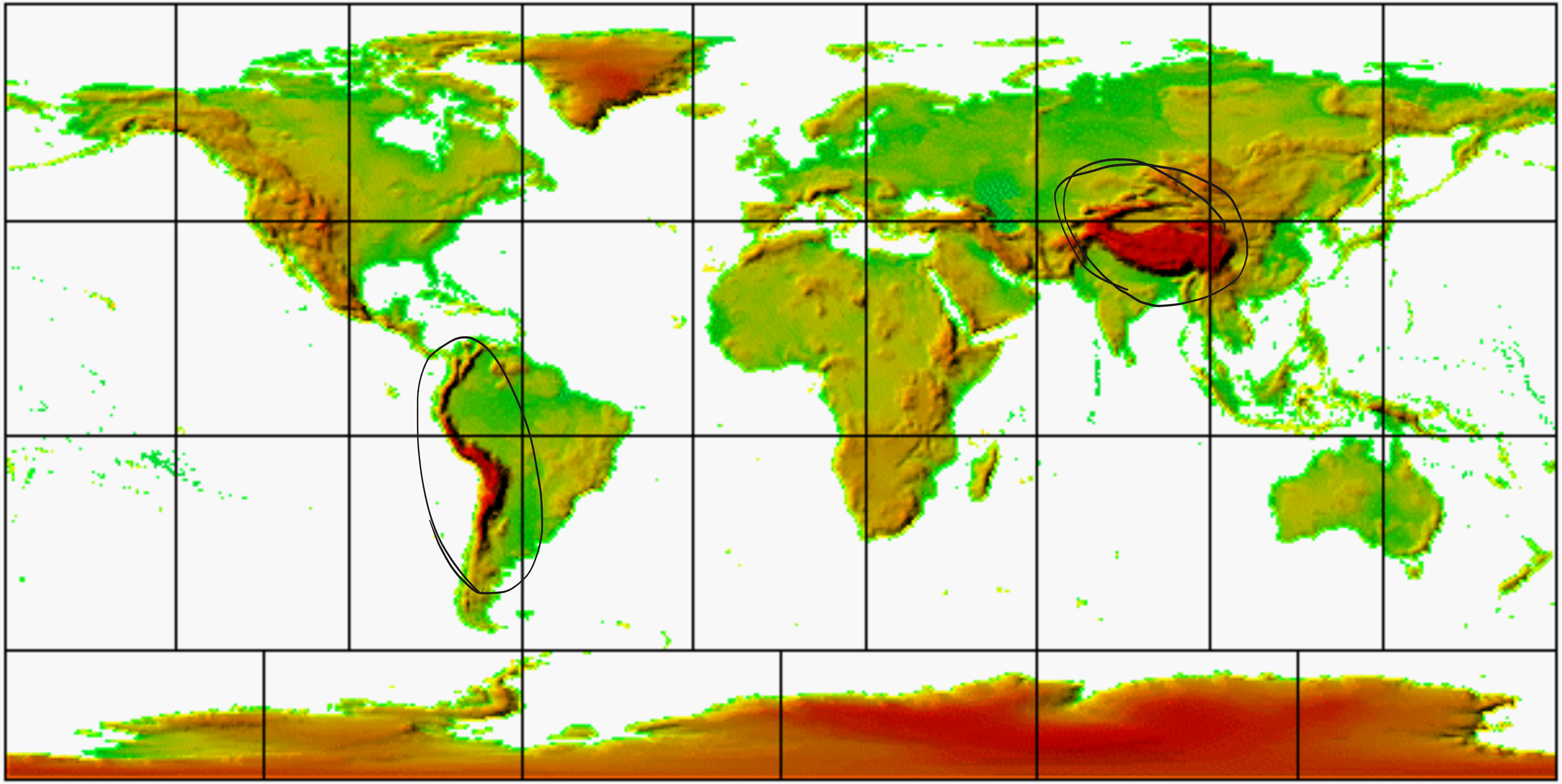
Contrasting results in ambiguous cases of specific adaptation

High altitude adaptation

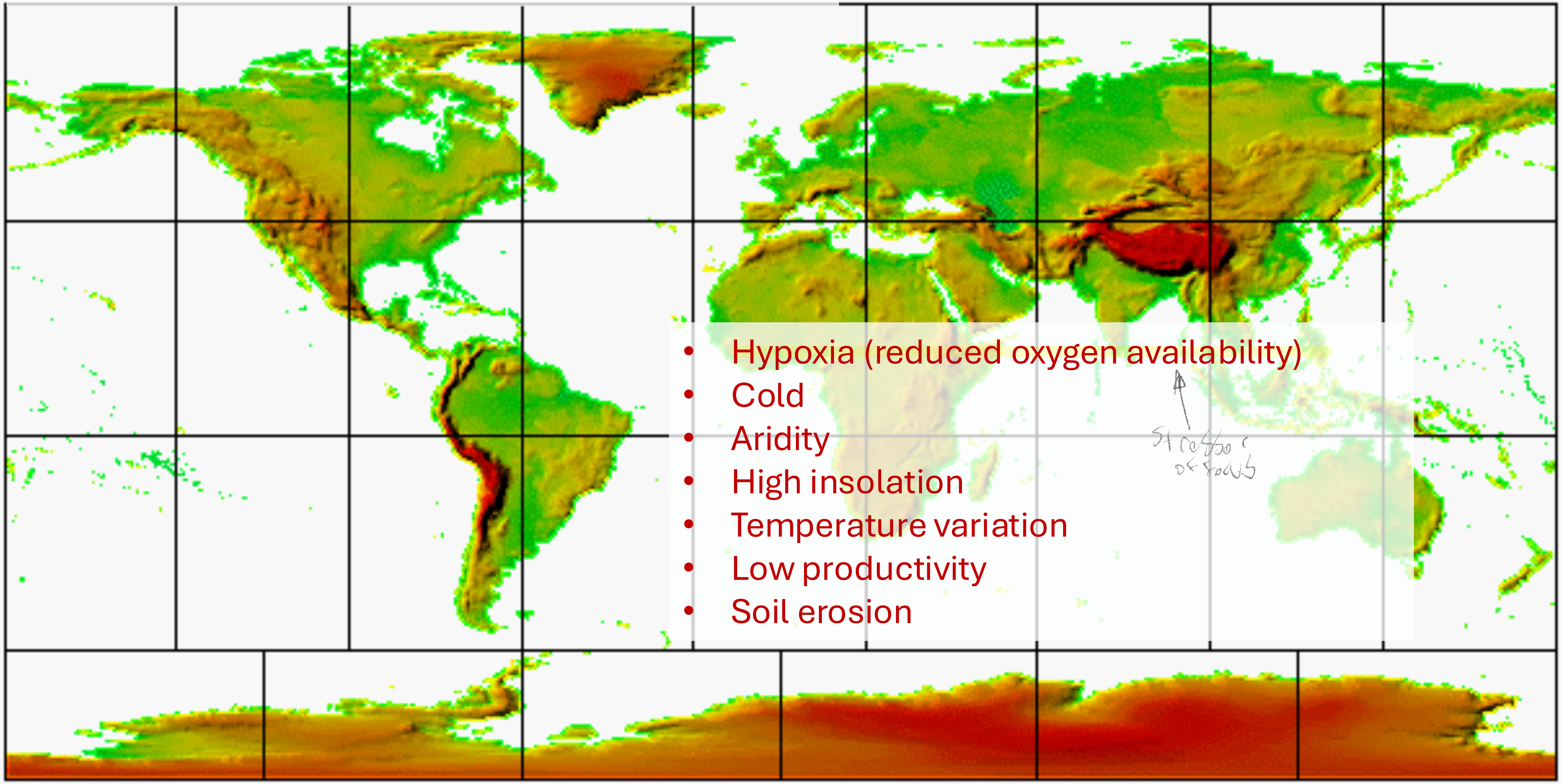
Dr. Laura van Holstein

lavanho@emory.edu

Room 215



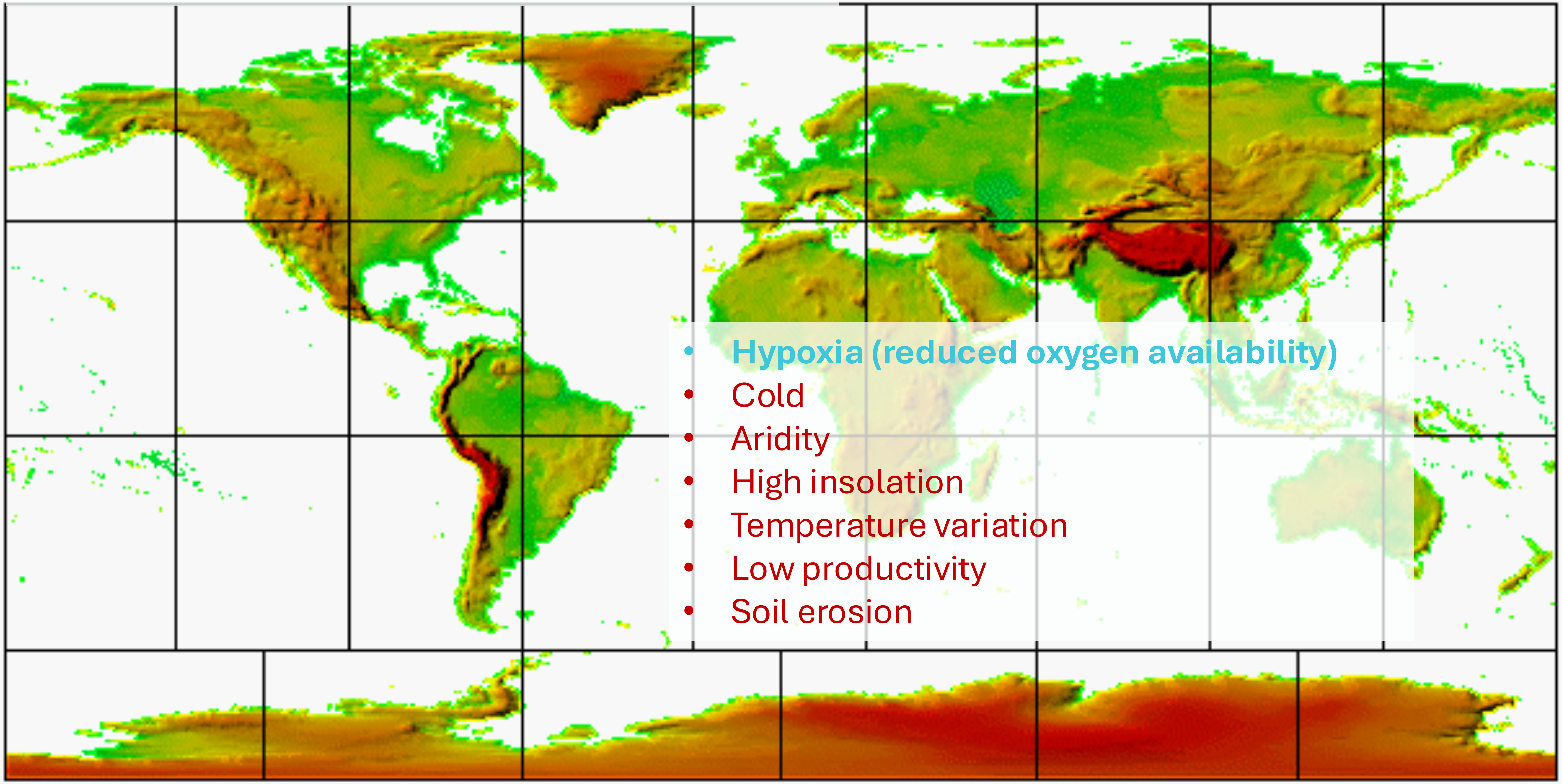
High altitude: many stressors



- Hypoxia (reduced oxygen availability)
- Cold
- Aridity
- High insolation
- Temperature variation
- Low productivity
- Soil erosion

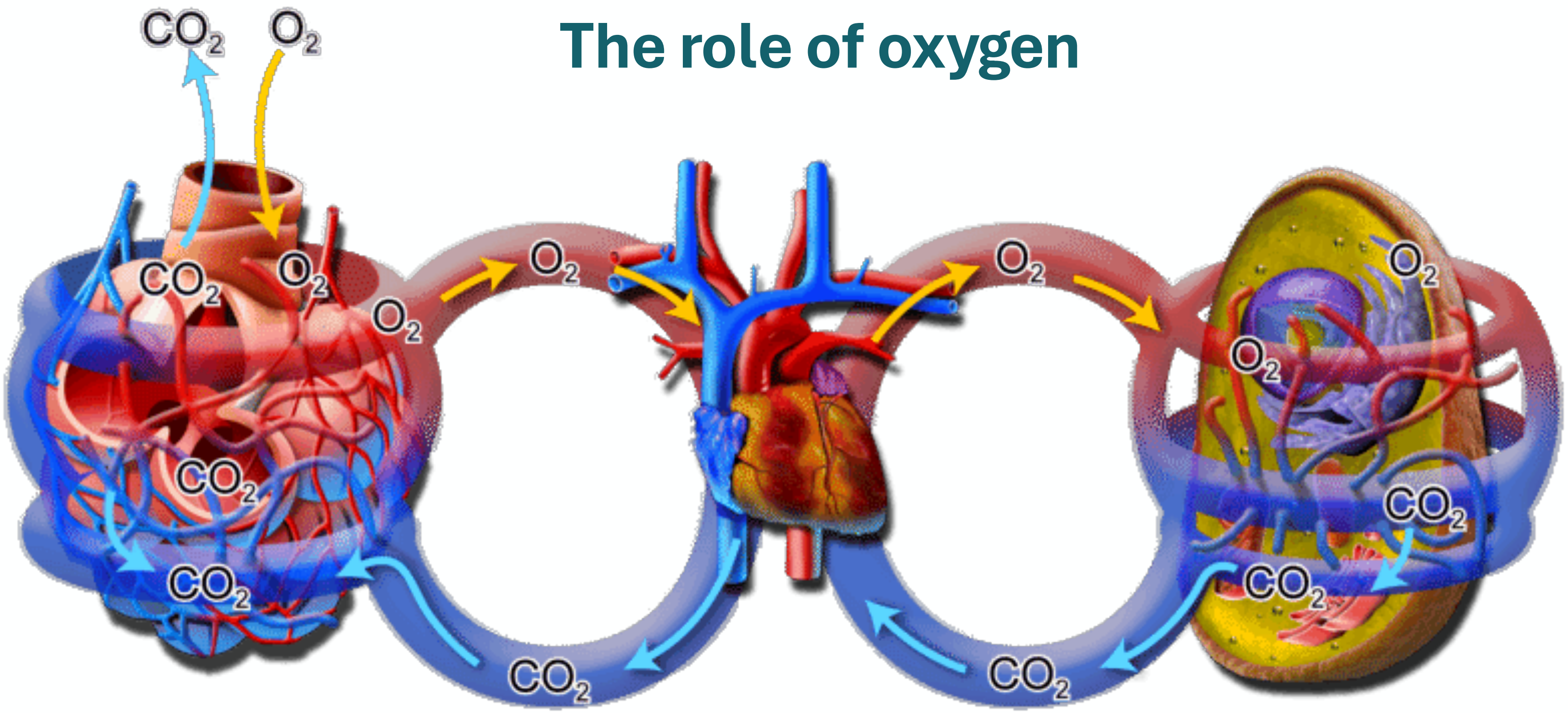
Stressor of focus

High altitude: many stressors

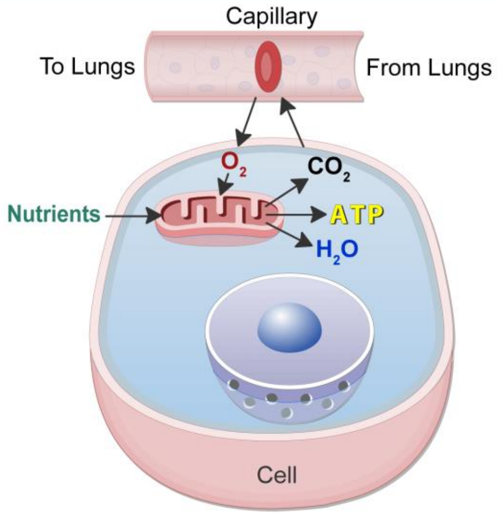


- Hypoxia (reduced oxygen availability)
- Cold
- Aridity
- High insolation
- Temperature variation
- Low productivity
- Soil erosion

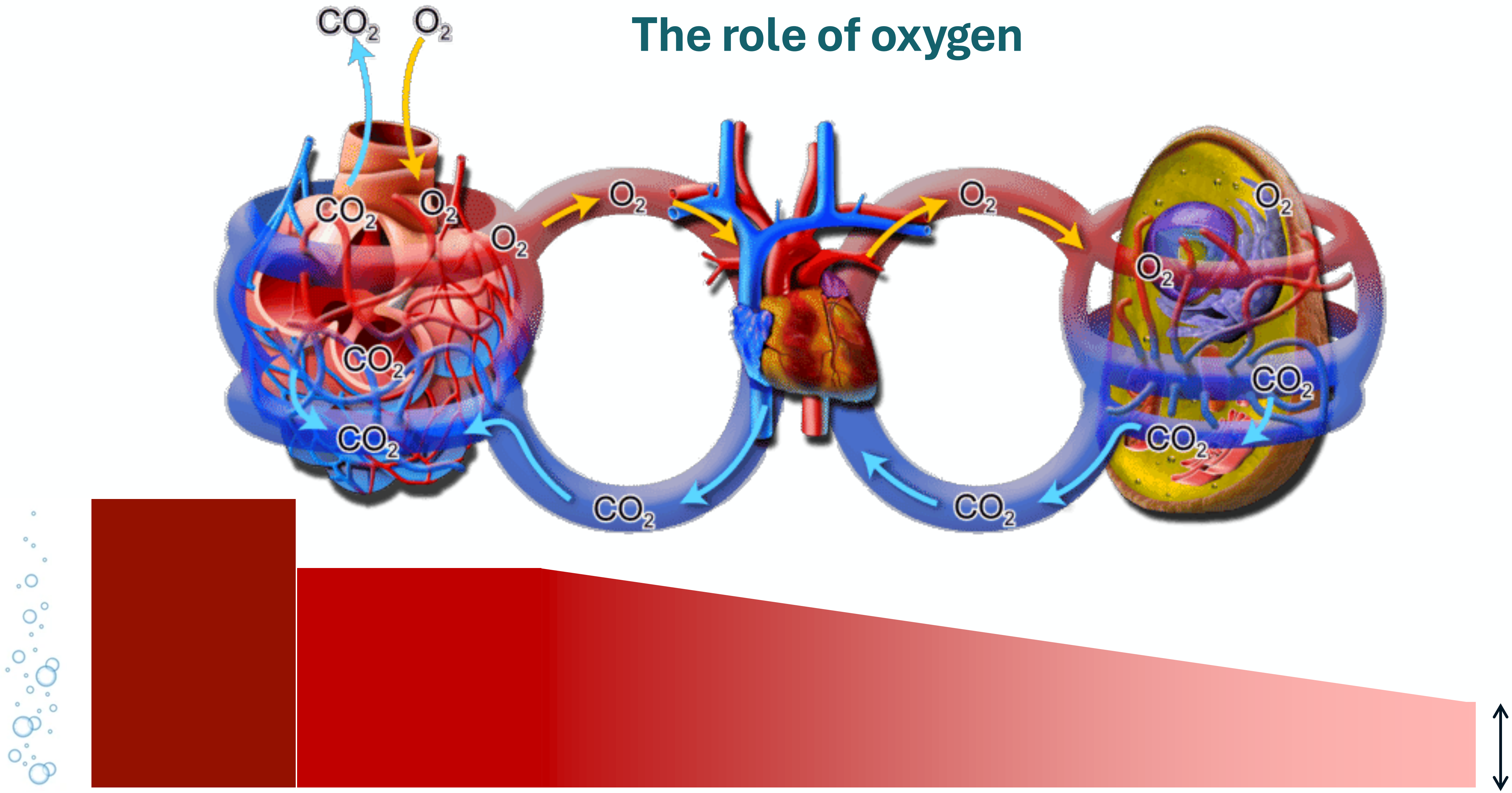
The role of oxygen



Alveolus surrounded by capillaries



The role of oxygen

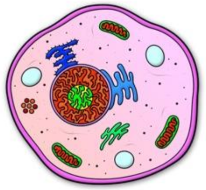


Ambient oxygen pressure (Ambient PO_2)

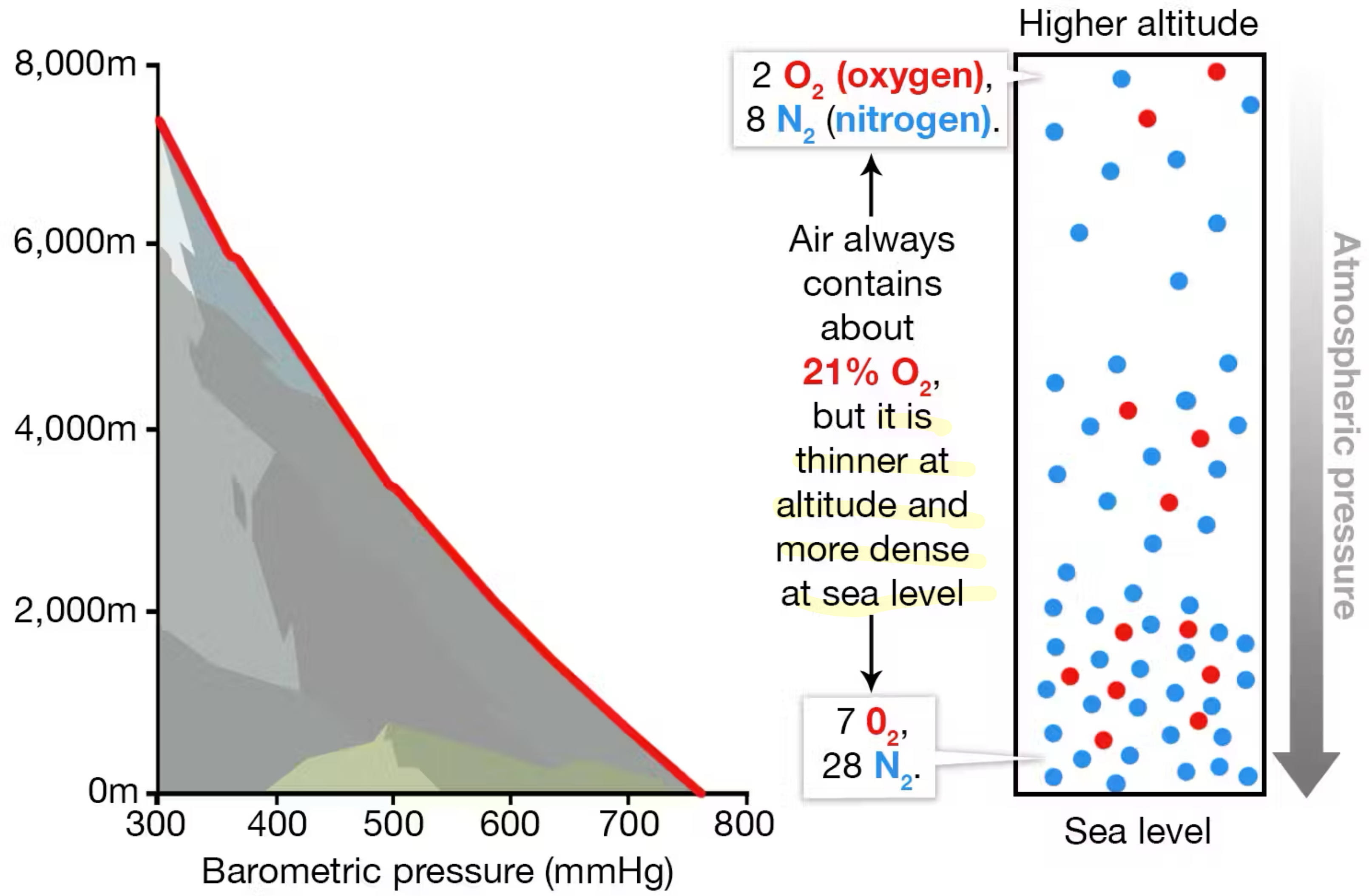
Alveolar oxygen pressure (P_AO_2)

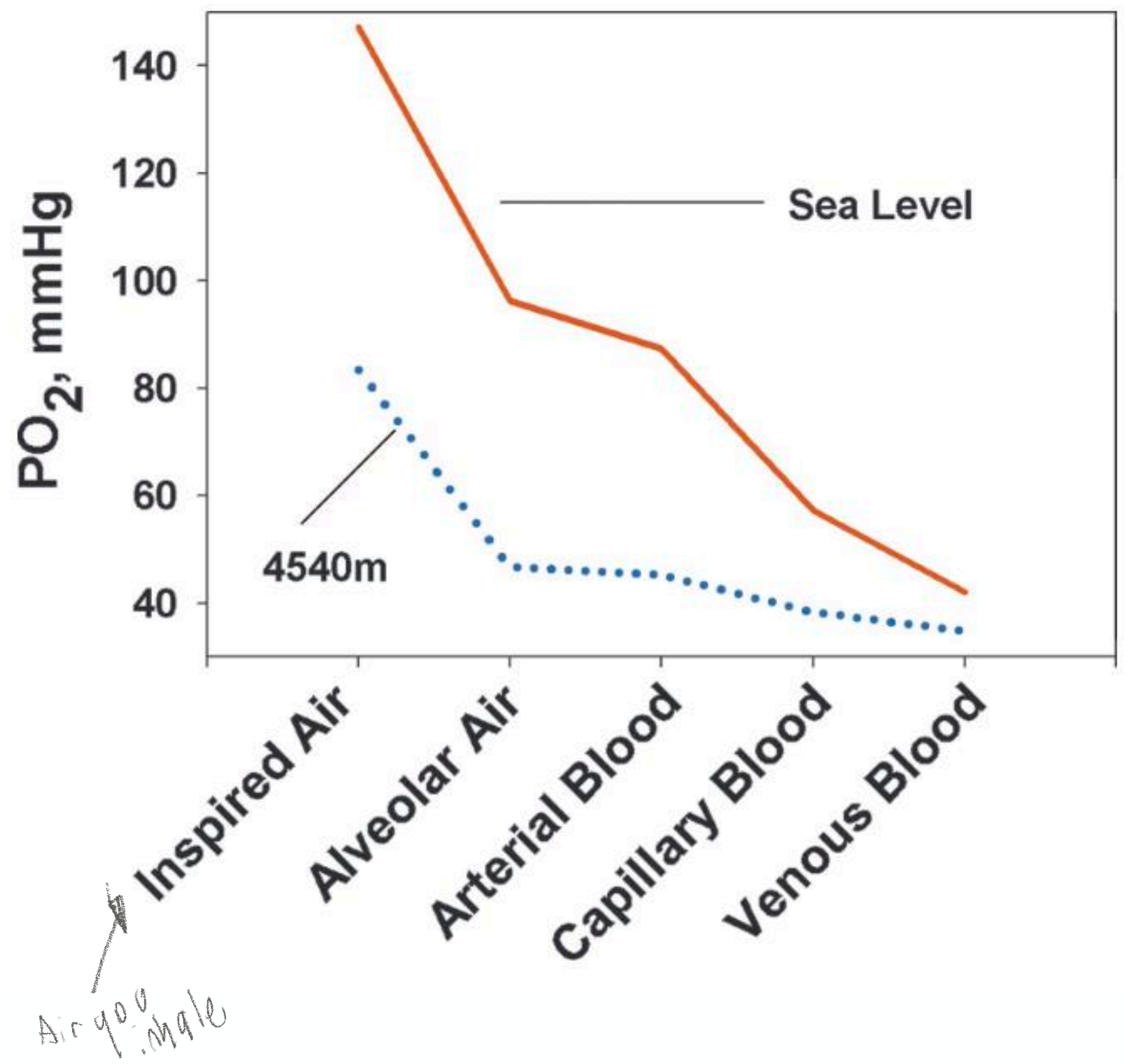
Saturation of haemoglobin (SaO_2)

O_2 content of blood per liter = SaO_2 + Hb + blood flow

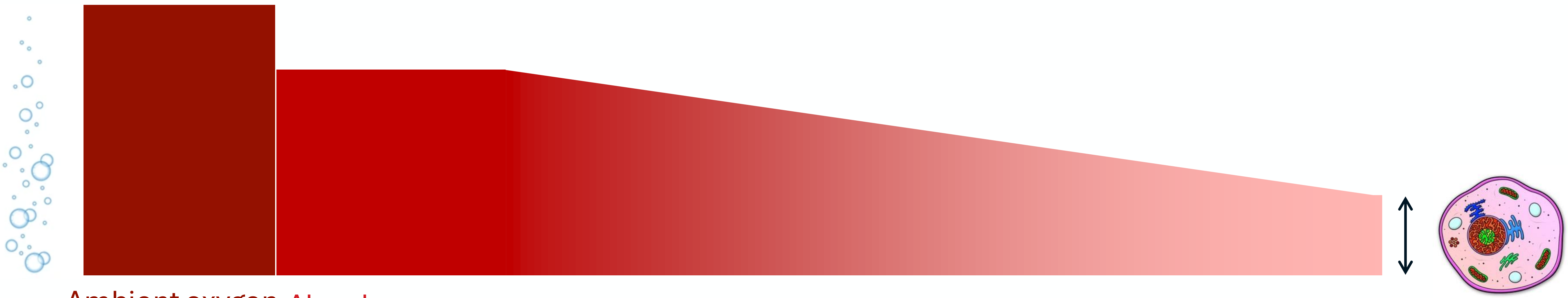


The impact of altitude on oxygen levels




PO2, mmHg
140
120
100
80
60
40
Sea Level
4540m
Inspired Air
Alveolar Air
Arterial Blood
Capillary Blood
Venous Blood
Air you inhale

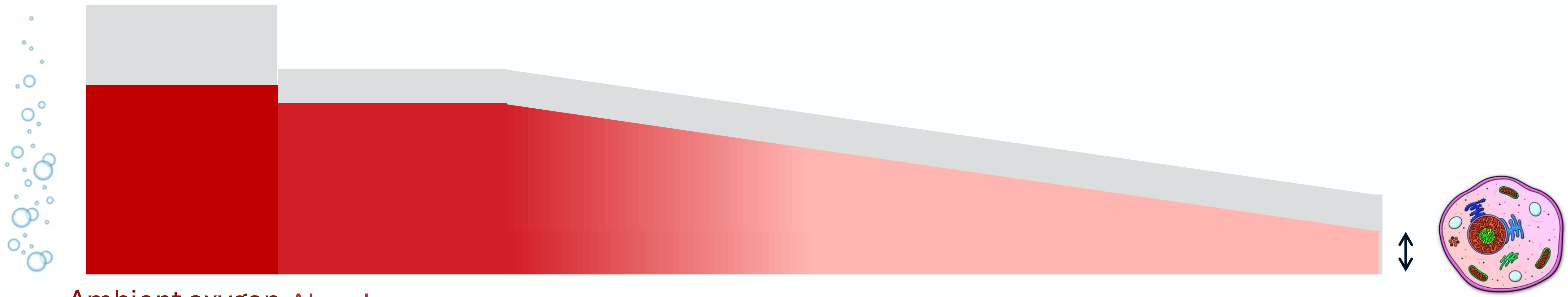
The role of oxygen



The role of oxygen

Less oxygen available and diffusion slows down due to smaller pressure differences

Under hypoxia: the pressure differences among different stages of oxygen transport are smaller: diffusion rate is lower.



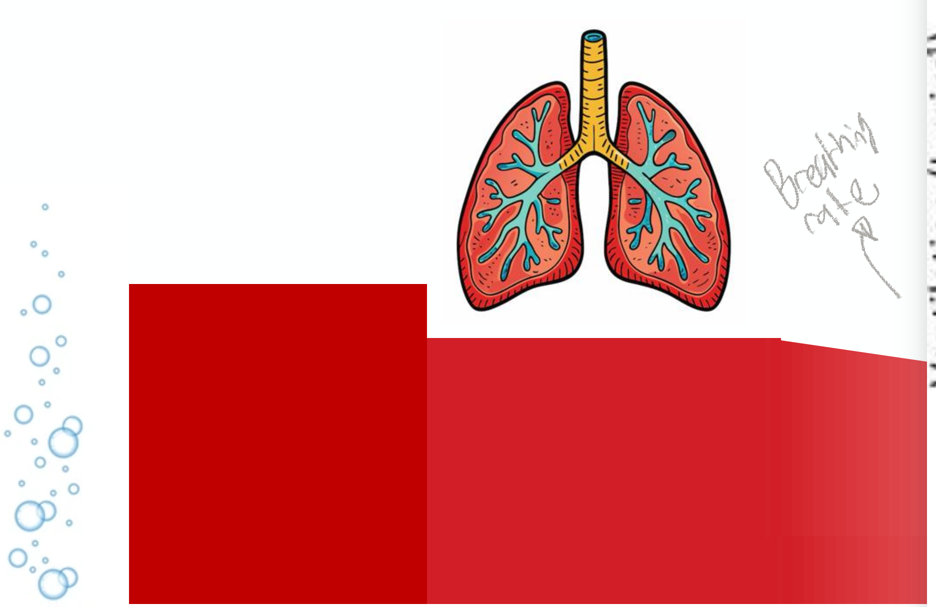
Ambient oxygen pressure (Ambient PO_2)

Alveolar oxygen pressure (P_AO_2)

Saturation of haemoglobin (SaO_2)
O_2 content of blood per liter = SaO_2 + Hb + blood flow

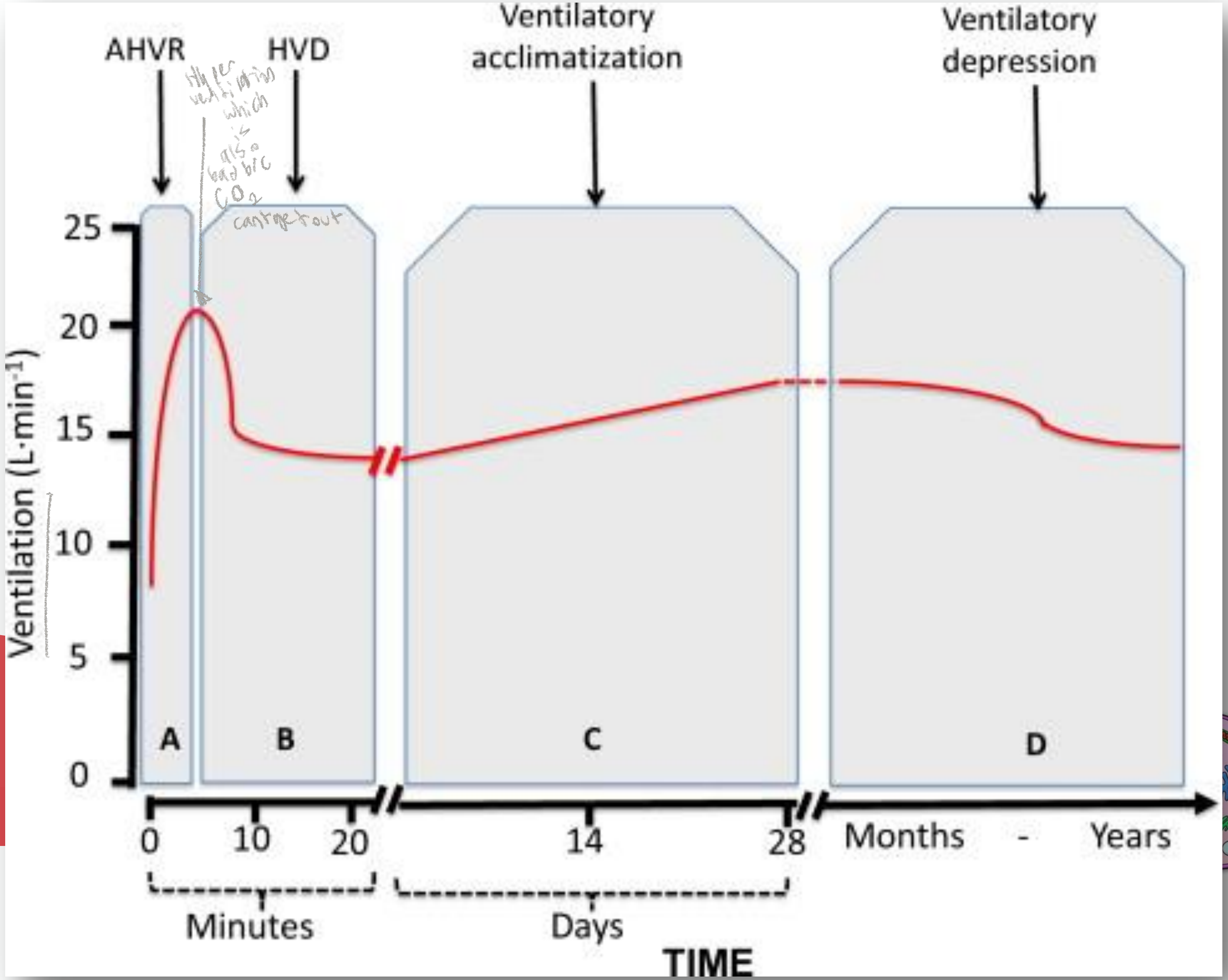
What now?!

Increase breathing rate



Ambient oxygen pressure (Ambient PO_2)

Alveolar oxygen pressure (P_AO_2)

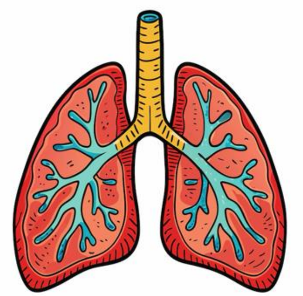


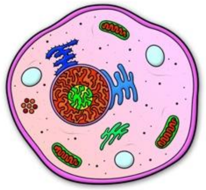
The problem
Your solutions
Natural selection's solutions

What now?!

Increase breathing rate

↓





Ambient oxygen pressure (Ambient PO_2)

Alveolar oxygen pressure (P_AO_2)

Saturation of haemoglobin (SaO_2)

Diff parts to act on to increase differential

# of RBCs

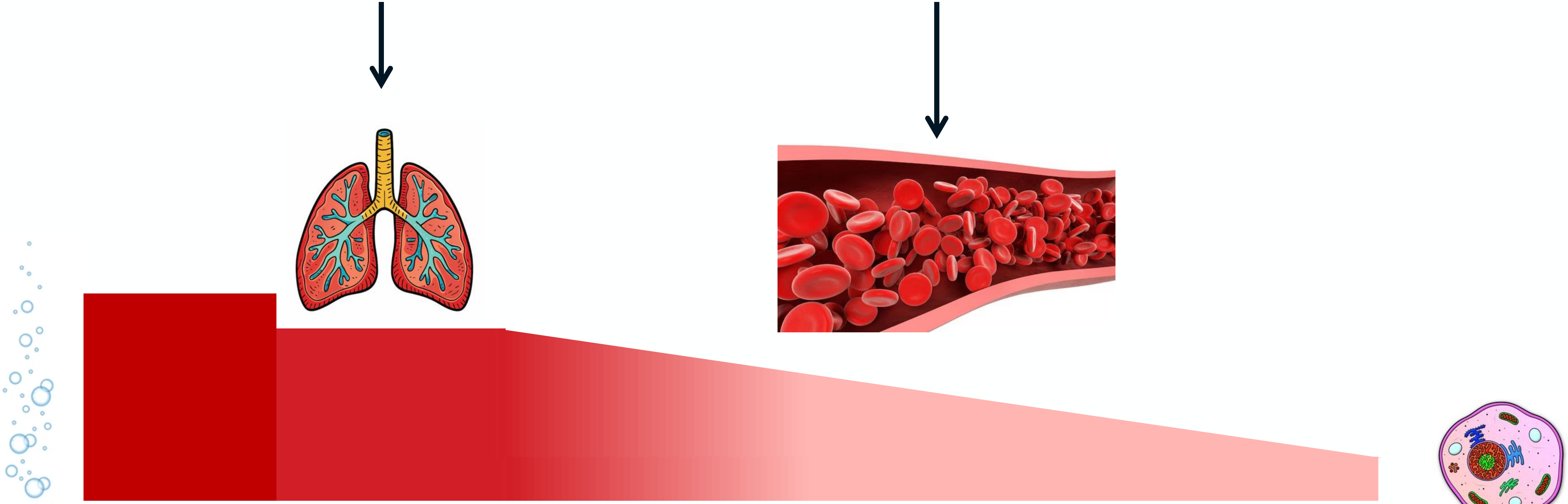
O_2 content of blood per liter = **SaO_2** + Hb + blood flow

↑ such as HR increase

What now?!

Increase breathing rate

Increase haemoglobin
Increase blood flow



Ambient oxygen pressure (Ambient PO_2)

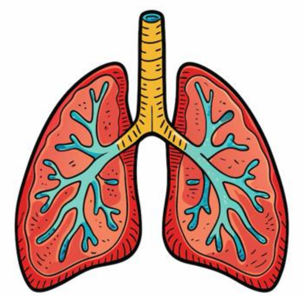
Alveolar oxygen pressure (P_AO_2)

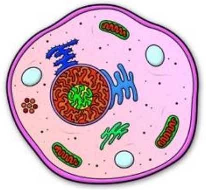
Saturation of haemoglobin (SaO_2)
O_2 content of blood per liter = SaO_2 + **Hb + blood flow**

What now?!

Increase breathing rate

Increase haemoglobin
Increase blood flow





Ambient oxygen pressure (Ambient PO_2)

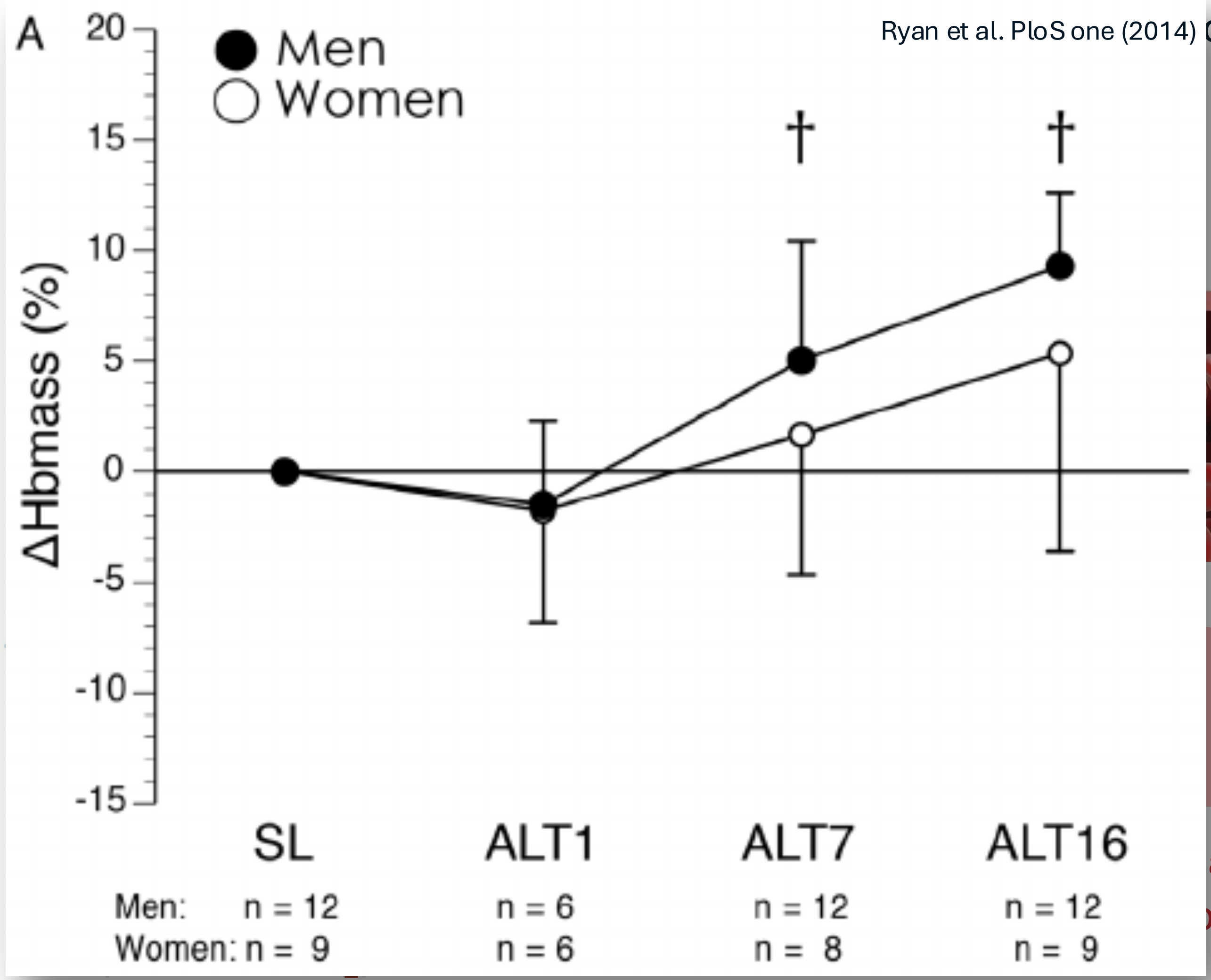
Alveolar oxygen pressure (P_AO_2)

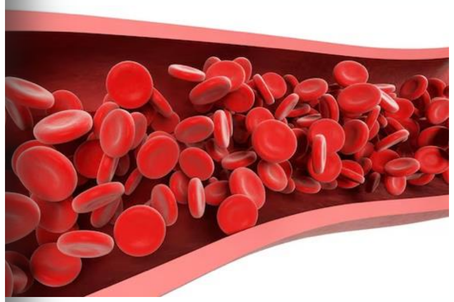
Saturation of haemoglobin (SaO_2)
O_2 content of blood per liter = SaO_2 + Hb + blood flow

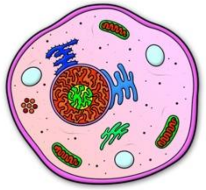
What now?!

Increase haemoglobin

crease blood flow







aemoglobin (SaO_2)

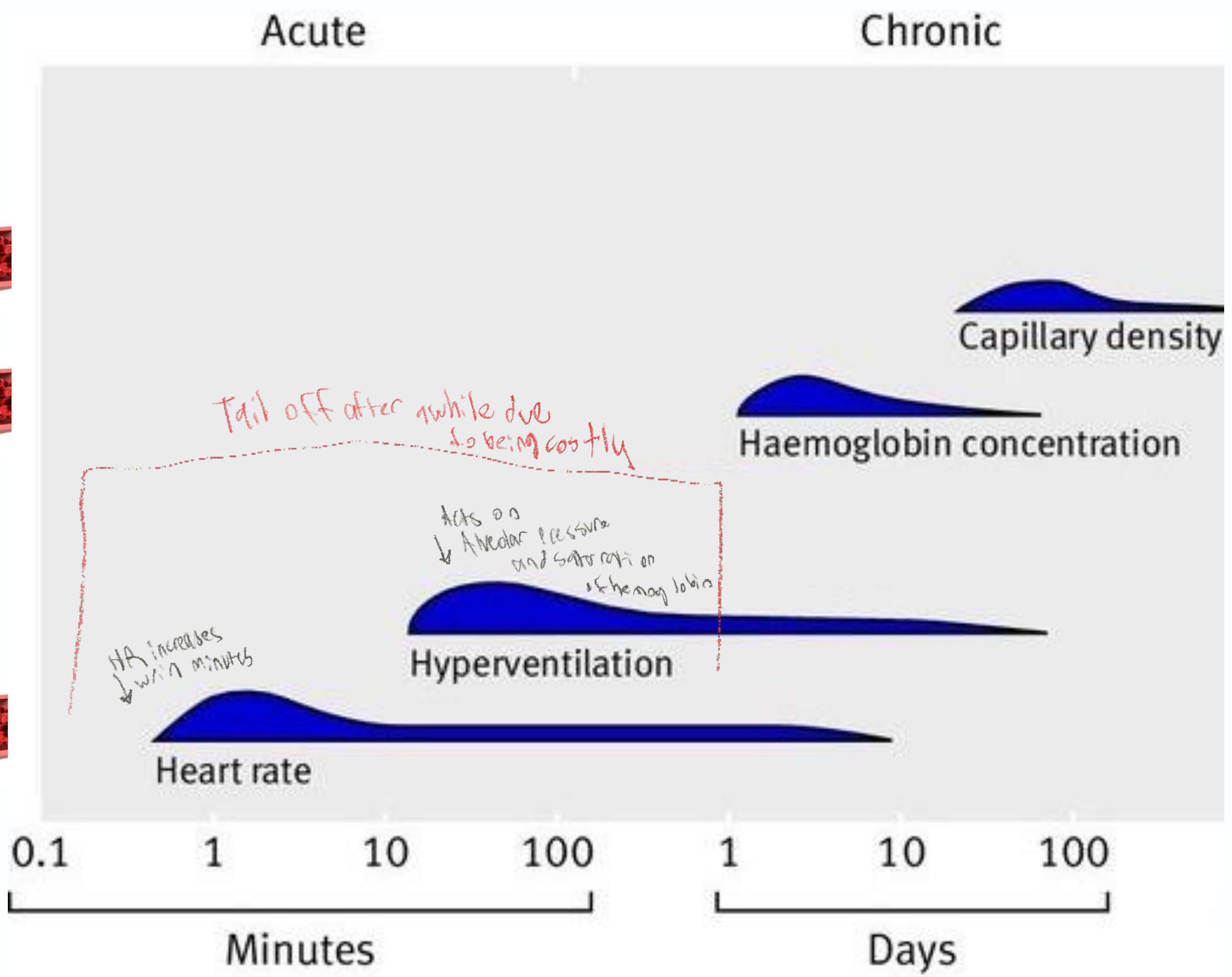
lood per liter = SaO_2 + Hb + blood flow

What now?!

Flow 

Hb 

Flow 



Hypoxia-inducible factors

Transcription factor
protein that
act on
expression
of genes

EGLN1

codes for PHD2

PHD2

PHD2
degrades HIFs
depending on
oxygen availability

Degradation, dependent on
oxygen availability

HIF1A

EPAS1 codes for HIF2A

HIF3A

Regulates
production
of

Protein helps form RBCs

Erythropoietin
(erythrocytosis)

Transferrin
(transfer of iron,
necessary for
haemoglobin)

Transcription factors

Target genes/proteins

Hypoxia-inducible factors

EGLN1

PHD2

High oxygen

HIF1A

EPAS1 → HIF2A

HIF3A

Erythropoietin
(erythrocytosis)

Transferrin
(transfer of iron, necessary for haemoglobin)

Transcription factors → *Target genes/proteins*

Hypoxia-inducible factors

EGLN1

PHD2

Low oxygen

Upregulates HIFs

HIF1A
HIF1A
HIF1A

EPAS1

HIF2A
HIF2A
HIF2A
HIF3A

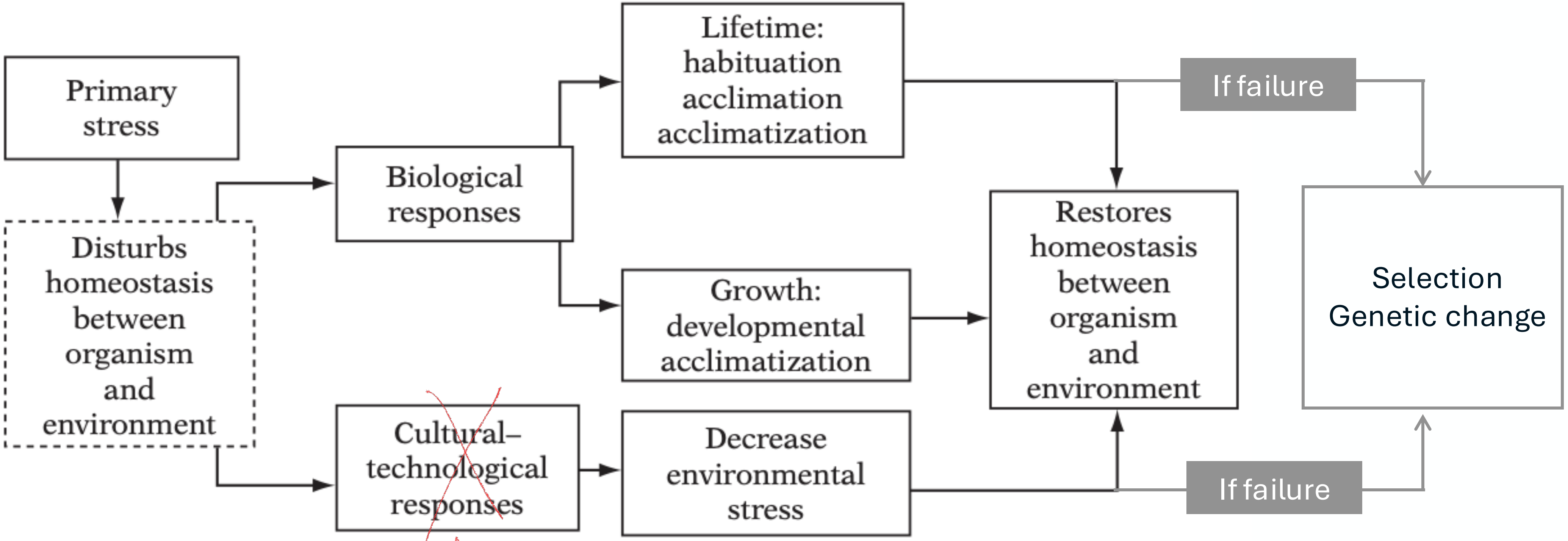
Transcription factors

Erythropoietin
(erythrocytosis)

Transferrin
(transfer of iron, necessary for haemoglobin)

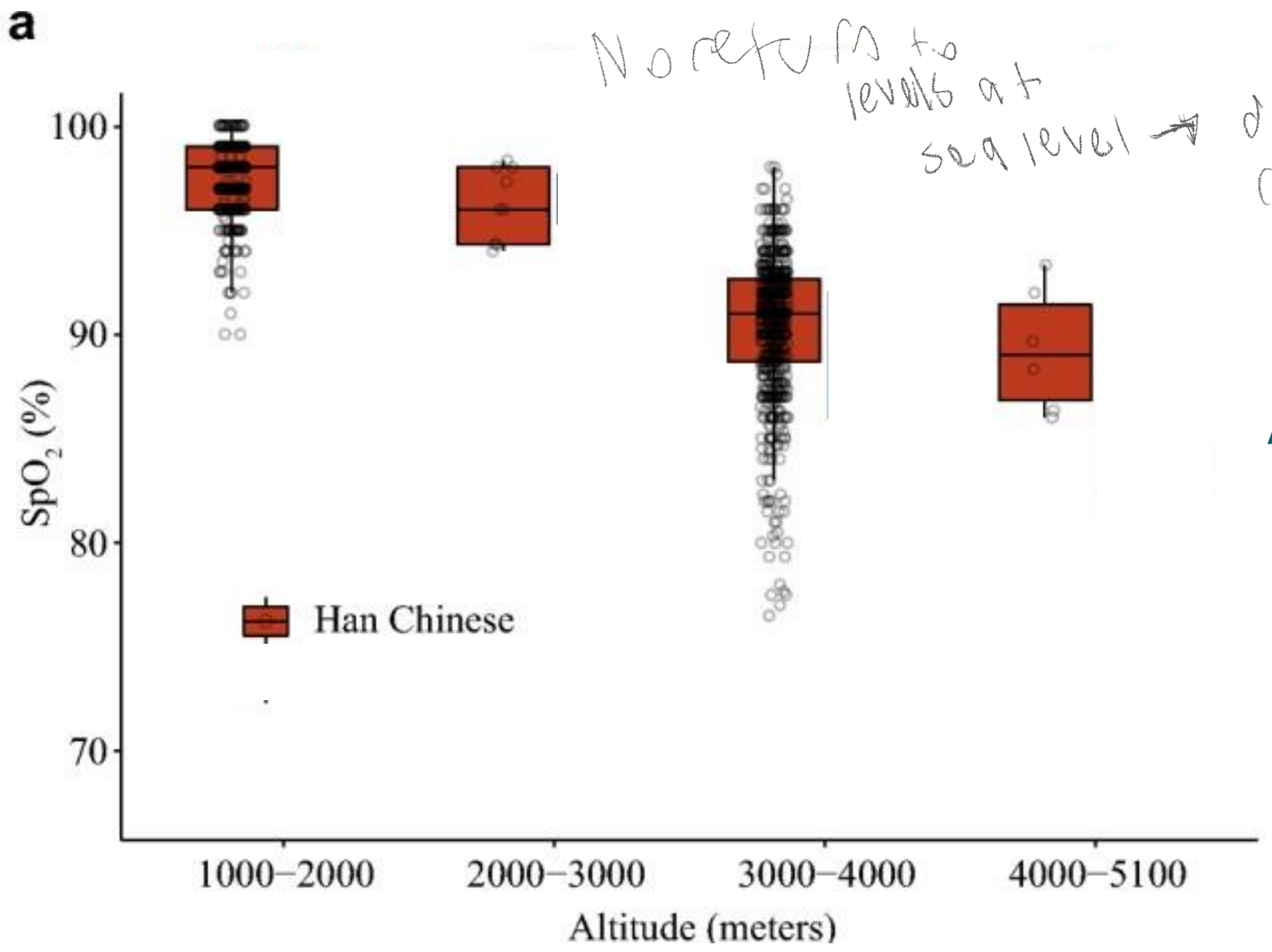
Target genes/proteins

Recall lecture 3



won't
help at high
altitudes are
for oxygen tank

Effectiveness & costs of these responses

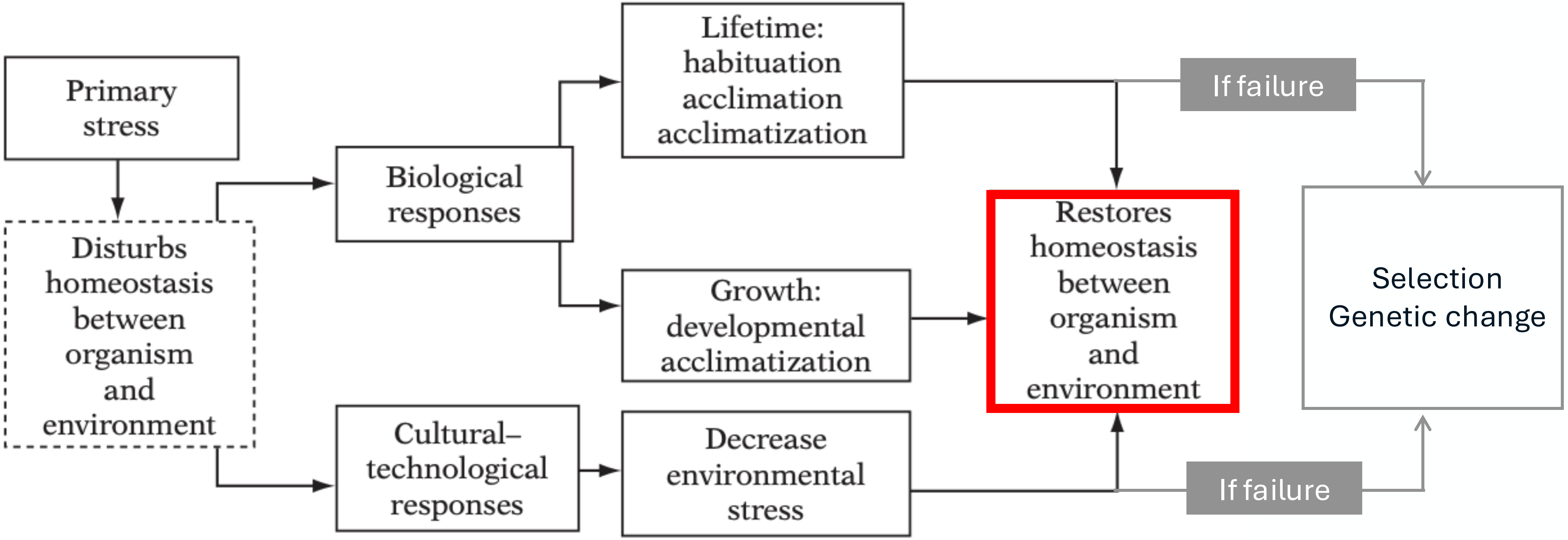


No return to levels at sea level → doesn't fully restore homeostasis

Acclimatized individuals

He et al. Phenomics (2023)

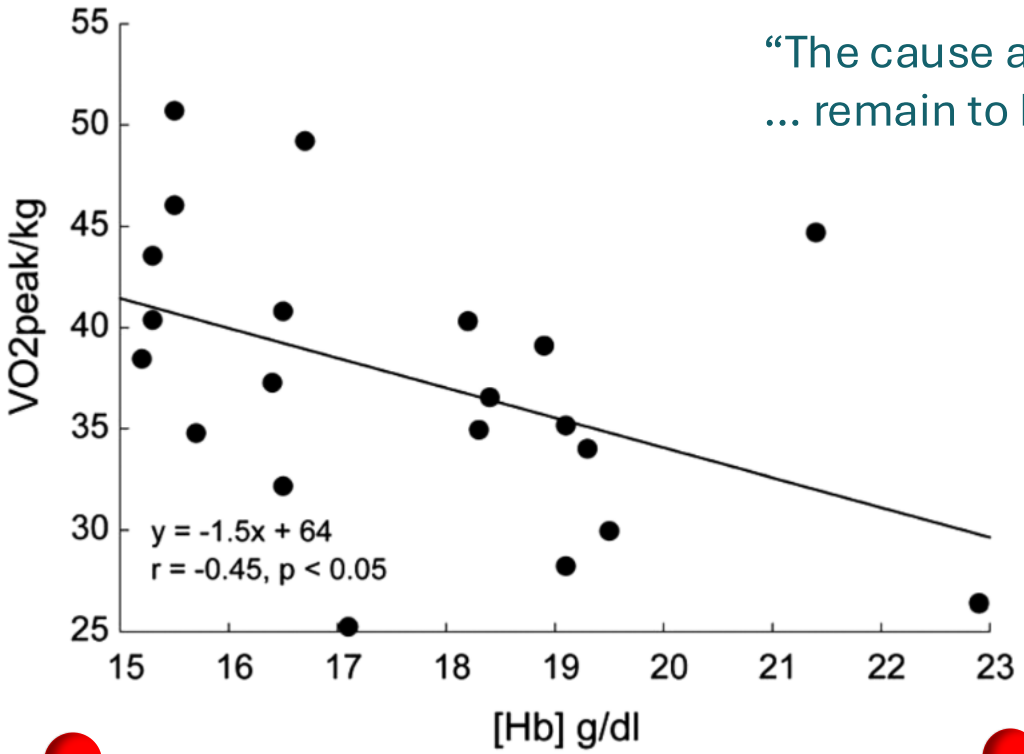
Recall lecture 3



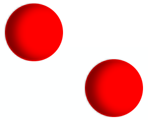
Effectiveness & costs of these responses

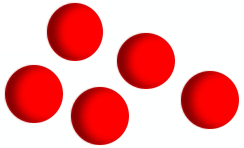
- Reduced peak exercise capacity

Peak exercise capacity



“The cause and effect relationships … remain to be elucidated.” (!)





Haemoglobin concentration

Simonson et al. J. Physiology (2015)

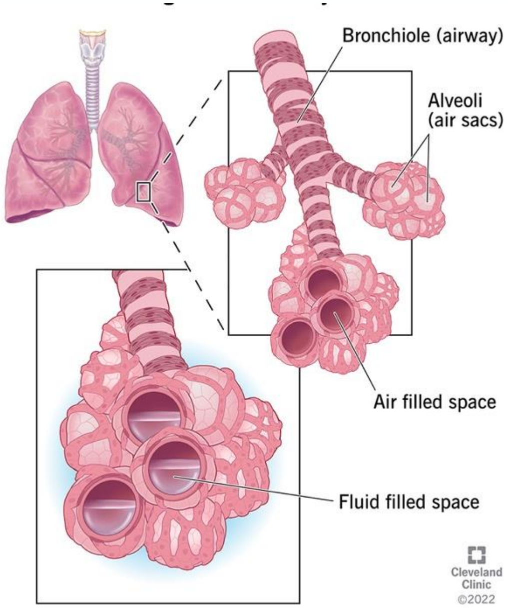
Effectiveness & costs of these responses

- Reduced peak exercise capacity
- Chronic Mountain Sickness; other cardiovascular diseases

Chronic Mountain Sickness

Excessive erythrocytosis:

- Hypertension
- Risk of pulmonary and cerebral oedema (fluid build-up in lungs and brain)
- Risk of stroke



Effectiveness & costs of these responses

- Reduced peak exercise capacity
- Chronic Mountain Sickness; other cardiovascular diseases
- Poorer reproductive outcomes

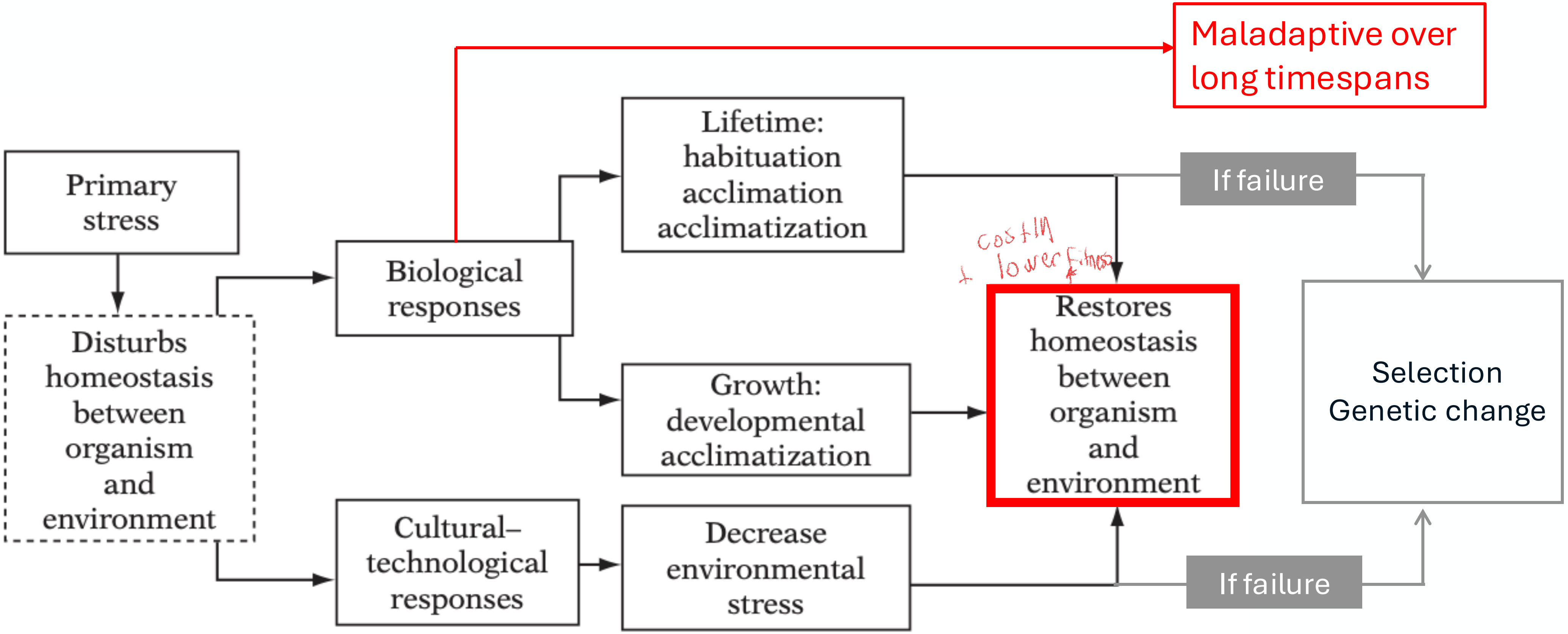
Continuous variable	N	Mean	SD	Min	Max
Pregnancies, #	1006	5.6	2.86	0	15
Live births, #	1006	5.4	2.77	0	14
Children Surviving to the age of 15 years, #	979	4.0	2.14	0	11
Age, years	1006	55	11	39	91
Age at first pregnancy and birth, years	982	24	5	15	46
Age at last pregnancy and birth*, years	982	37	6	19	58
Hemoglobin concentration, gm/dl	959	13.8	1.51	5.2	18.7
O2 Saturation, %	959	87.4	4.5	60	96
Pulse, f/min	958	74	11	49	121

Categorical variable		Count (%)	# Missing (%)
Current marital status	1 Married	659 (65.51)	
	2 Widowed	252 (25.05)	
	3 Divorced/separate	40 (3.98)	
	4 Never married	55 (5.47)	
Use of contraception	0 Never	682 (67.79)	10 (0.99)
	1 Past use	147 (14.61)	
	2 Current	167 (16.6)	
District of residence	1 Gorkha	264 (26.24)	
	2 Mustang	742 (73.76)	
Type of marriage	0 Never	55 (5.47)	
	1 Cousin, not polyandrous	72 (7.16)	
	2 Not cousin, is polyandrous	94 (9.34)	
	3 Cousin and polyandrous	16 (1.59)	
	4 Not cousin, not polyandrous	769 (76.44)	

Lower haemoglobin content:
better reproductive outcomes

Cho et al. Evolution, Medicine, and Public Health (2017)

Recall lecture 3



The problem
Your solutions
Natural selection's solutions

Adaptive responses to high-altitude hypoxia

↑ Increase; costly

Homeostasis + acclimation

	Sojourners
Resting ventilation	↑
Hb concentration	↑
Blood flow	↑
Capillary density	↑

people born at low altitudes who go to high altitudes

Prevent short-term risk to life...

But at **high long-term cost**

Birth weight Offspring survival	↓

And not even that effective...

Adaptive responses to high-altitude hypoxia

↑ Increase; costly

Homeostasis + acclimation

	Sojourners
Resting ventilation	↑
Hb concentration	↑
Blood flow	↑
Capillary density	↑

Birth weight Offspring survival	↓

A 50-year period of infertility in early Spaniards in the Andes at 4100m!

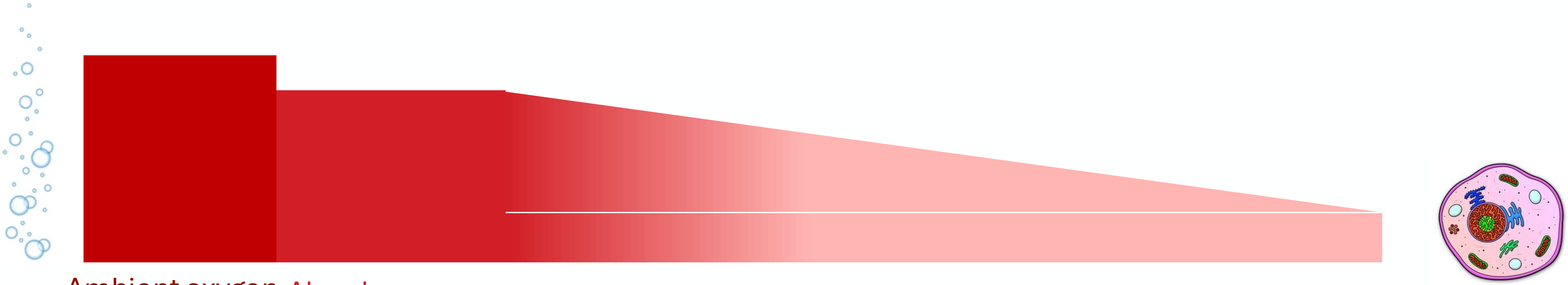
Preexisting Medical Conditions

People with pre-existing medical conditions should talk with a doctor before traveling to high elevation.

- People with heart or lung disease should talk to a doctor who is familiar with high-altitude medicine before their trip.
- People with diabetes need to be aware that their illness may be difficult to manage at high elevation.
- Pregnant women can make brief trips to high elevations but they should talk with their doctor because they may be advised not to sleep at elevations above 10,000 feet.
- People with some illnesses (e.g., sickle cell anemia, severe pulmonary hypertension) should not travel to high elevations under any circumstances.

Fitness costs: natural selection

With falling barometric pressure comes rising evolutionary pressure



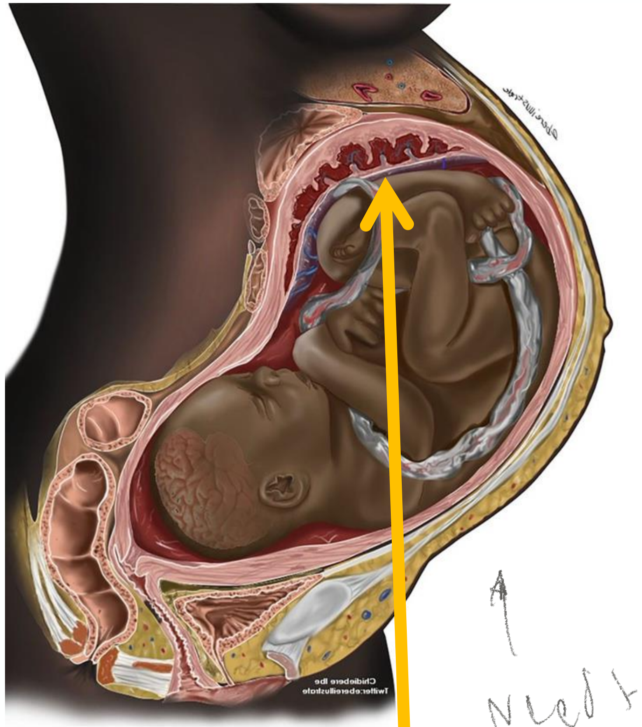
Ambient oxygen pressure (Ambient PO_2)

Alveolar oxygen pressure (P_AO_2)

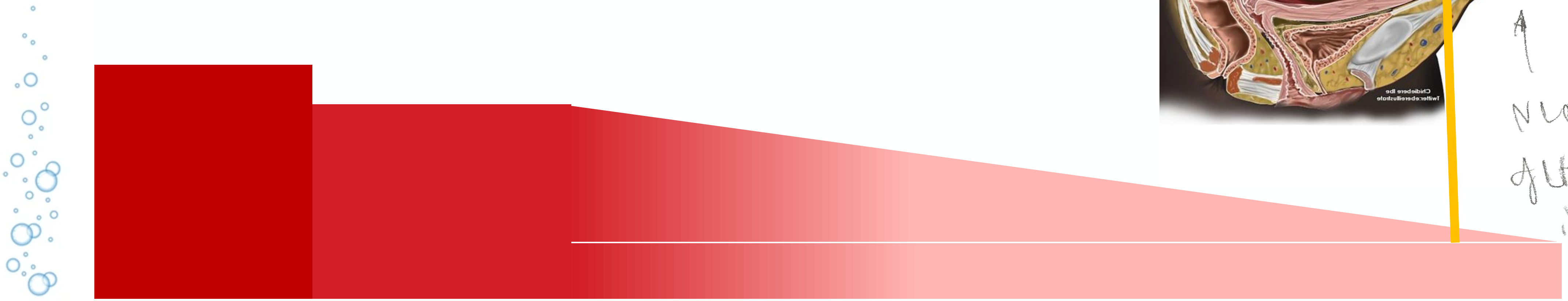
Saturation of haemoglobin (SaO_2)

O_2 content of blood per liter = SaO_2 + Hb + blood flow

With falling barometric pressure comes rising evolutionary pressure



↑ Need to get oxygen into developing fetus

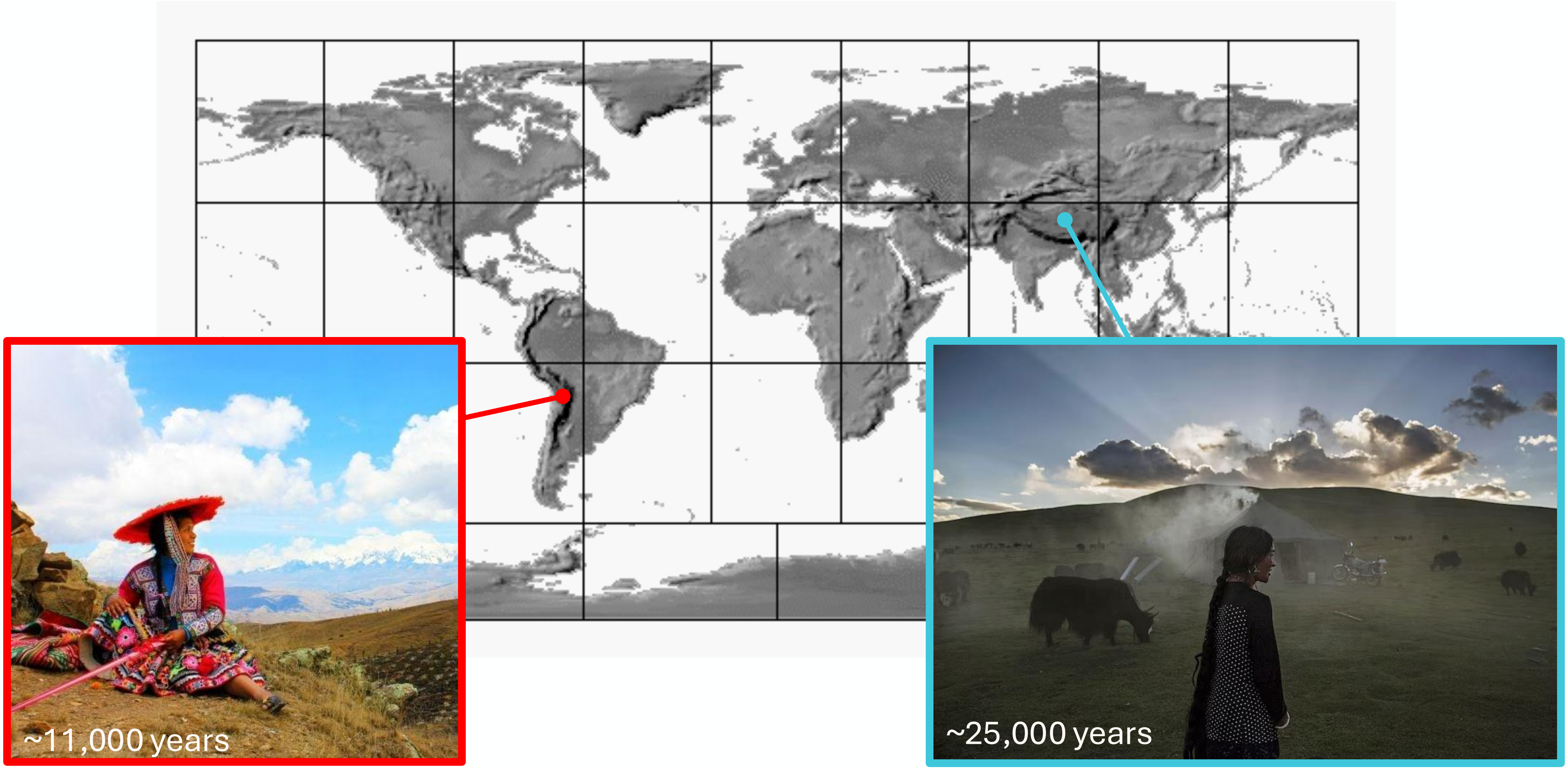


Ambient oxygen pressure (Ambient PO_2)

Alveolar oxygen pressure (P_AO_2)

Saturation of haemoglobin (SaO_2)

O_2 content of blood per liter = SaO_2 + Hb + blood flow


~11,000 years
~25,000 years

Adaptive responses to high-altitude hypoxia

Homeostasis + acclimation

	Sojourners	Tibetans	Andeans
Resting ventilation	↑		
Hb concentration	↑		
Blood flow	↑		
Capillary density	↑		
Birth weight Offspring survival	↓	—	—

Adaptive responses to high-altitude hypoxia

Homeostasis + acclimation

Same stress in 2 different pops – Tibetans mix w/ Denisovans who are likely adapted + Tibetans have more time → local adaptation takes a long time to evolve adaptive responses

High altitude is one of the strongest selective pressures on genome along w/ disease

tension btwn immediate survival but there are lifelong negative effects → otherwise forgo these responses + come up w something else

	Sojourners	Tibetans	Andeans
Resting ventilation	↑		
Hb concentration	↑		
Blood flow	↑		
Capillary density	↑		
Birth weight Offspring survival	↓	—	—

You have been assigned to one of these two populations. **Evolve, over the course of the appropriate generations, the ability to have higher birthweight than sojourners.**

You start with the same acclimatizing responses as sojourners – which are driven by the HIF pathway, and which are, clearly, maladaptive *over the course of a lifetime*.

How do you counter these? Do you...

- suppress the hypoxic response and evolve other ways to increase oxygen delivery to your tissues and those of the foetus?
- permit the hypoxic responses and evolve adaptations to counter the evolutionary costs of these responses?
- find some other solution?

Think abt tradeoffs for different routes

Adaptive responses to high-altitude hypoxia

Homeostasis + acclimation

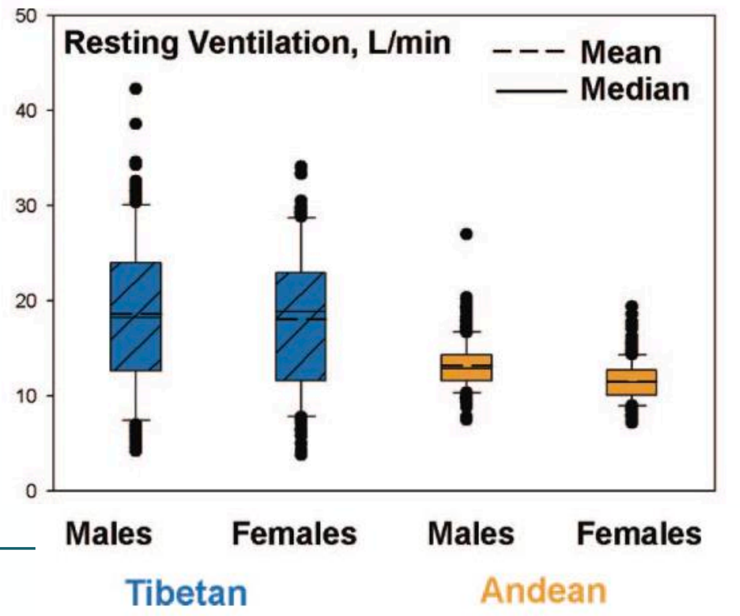
	Sojourners	Tibetans	Andeans
Resting ventilation	↑		
Hb concentration	↑		
Blood flow	↑		
Capillary density	↑		
Birth weight Offspring survival	↓	—	—

Adaptive responses to high-altitude hypoxia

Homeostasis + acclimation

— Means same as baseline at sea level for people born at sea level while sea level natives are at sea level

	Sojourners	Tibetans	Andeans
Resting ventilation	↑	↑	—
Hb concentration	↑		
Blood flow	↑		
Capillary density	↑		
Birth weight Offspring survival	↓		

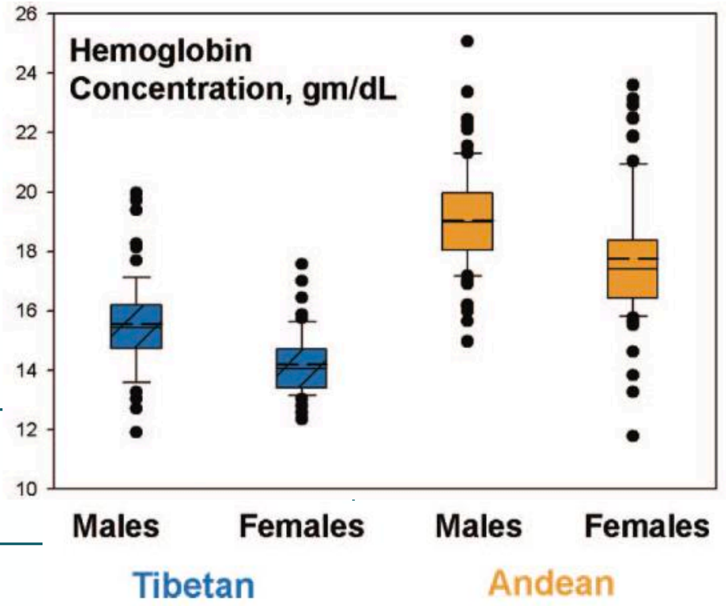


Beall PNAS 2007

Adaptive responses to high-altitude hypoxia

Homeostasis + acclimation

	Sojourners	Tibetans	Andeans
Resting ventilation	↑	↑	—
Hb concentration	↑	—	↑
Blood flow	↑		
Capillary density	↑		
Birth weight Offspring survival	↓		



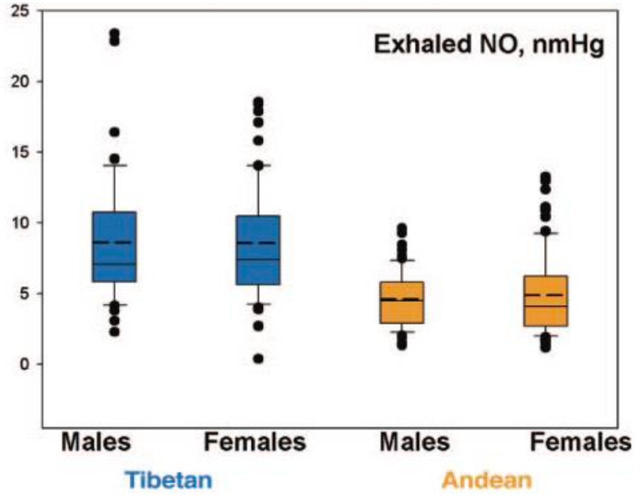
Beall PNAS 2007

The problem
Your solutions
Natural selection's solutions

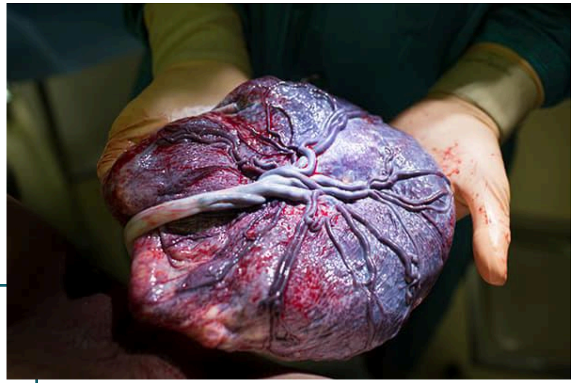
Adaptive responses to high-altitude hypoxia

Homeostasis + acclimation

	Sojourners	Tibetans	Andeans
Resting ventilation	↑	↑	—
Hb concentration	↑	—	↑
Blood flow	↑	↑ ↑	↓
Capillary density	↑		
Birth weight Offspring survival	↓		



Tibetan placentas: larger arteries, permitting higher blood flow

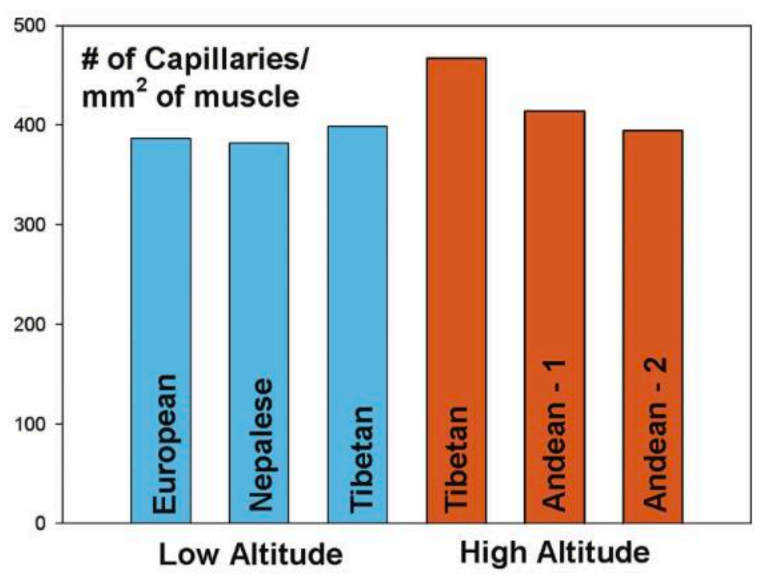


Beall PNAS 2007

Adaptive responses to high-altitude hypoxia

Homeostasis + acclimation

	Sojourners	Tibetans	Andeans
Resting ventilation	↑	↑	—
Hb concentration	↑	—	↑
Blood flow	↑	↑ ↑	↓
Capillary density	↑	↑ ↑	↑
Birth weight Offspring survival	↓	—	—



Beall PNAS 2007

Adaptive responses to high-altitude hypoxia

Homeostasis + acclimation

	Sojourners	Tibetans	Andeans
Resting ventilation	↑	↑	—
Hb concentration	↑	—	↑
Blood flow	↑	↑ ↑	↓
Capillary density	↑	↑ ↑	↑
Stature			
Lung volume			

Birth weight Offspring survival	↓	—	—

Beall PNAS 2007

Adaptive responses to high-altitude hypoxia

Homeostasis + acclimation

	Sojourners	Tibetans	Andeans
Resting ventilation	↑	↑	—
Hb concentration	↑	—	↑
Blood flow	↑	↑ ↑	↓
Capillary density	↑	↑ ↑	↑
Stature	—	↓	↓
Lung volume	—	↑	↑
Birth weight Offspring survival	↓	—	—

Beall PNAS 2007

Adaptive responses to high-altitude hypoxia

Homeostasis + acclimation

	Sojourners	Tibetans	Andeans
Resting ventilation	↑	↑	—
Hb concentration	↑	—	↑
Blood flow	↑	↑ ↑	↓
Capillary density	↑	↑ ↑	↑
Stature		↓	↓
Lung volume		↑	↑

Birth weight Offspring survival	↓	—	—

Tibetans:

Dampened erythrocytosis acclimation response

vs

Strategy based on blood flow

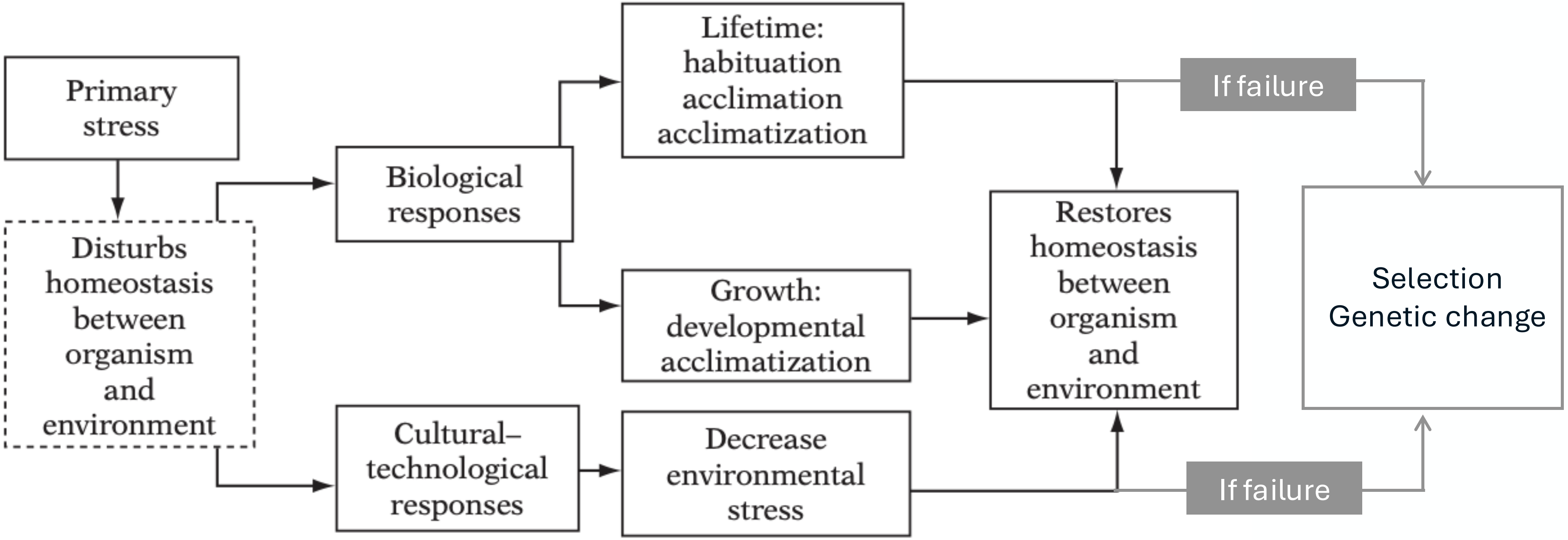
Andeans:

Strategy based on erythrocytosis

Not enough time to evolve other responses

Both: changed growth patterns

Recall lecture 3



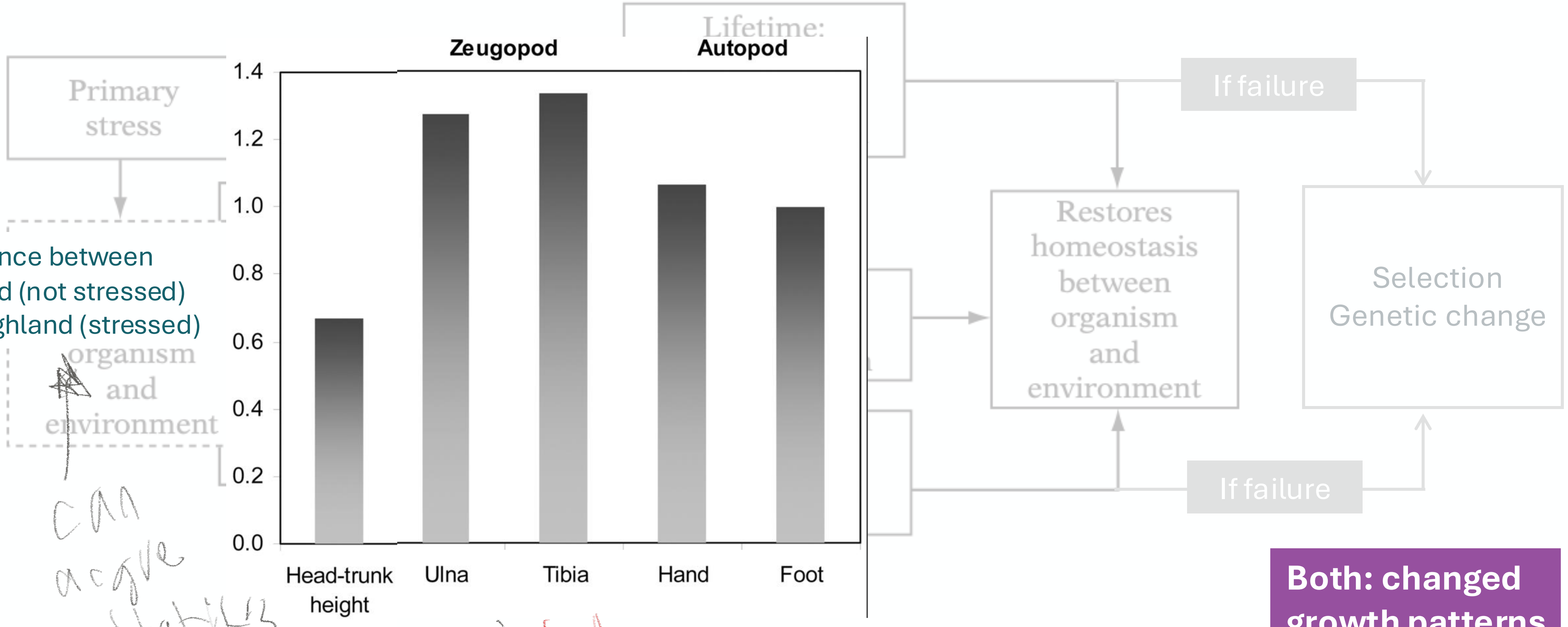
Both: changed growth patterns

Hard to tease apart genetic and developmental adaptation

Open question: is this genetic (reflecting long-term adaptation through natural selection) or developmental plasticity?

Recall lecture 3

Difference between lowland (not stressed) and highland (stressed)



can argue that it's developmental but...

Both: changed growth patterns

Open question: is this genetic (reflecting long-term adaptation through natural selection) or **developmental plasticity**?

Pomeroy et al. (2012) *PLoS ONE*

Recall lecture 3

Primary stress

Lifetime: habituation acclimation acclimatization

If failure

Selection
Genetic change

If failure

There is genetic evidence for selection on in Andeans on growth

Article

A positively selected *FBN1* missense variant reduces height in Peruvian individuals

https://doi.org/10.1038/s41586-020-2302-0

Received: 28 February 2019

Accepted: 10 March 2020

Published online: 13 May 2020

Check for updates

Samira Asgari[1,2,3,4,5], Yang Luo[1,2,3,4,5], Ali Akbari[4,6], Gillian M. Belbin[7,8,9], Xinyi Li[1,2,3,4,5], Daniel N. Harris[10,11], Martin Selig[12], Eric Bartell[4,5,13], Roger Calderon[14], Kamil Slowikowski[1,2,3,4,5], Carmen Contreras[14], Rosa Yataco[14], Jerome T. Galea[15], Judith Jimenez[14], Julia M. Coit[16], Chandel Farroñay[14], Rosalynn M. Nazarian[12], Timothy D. O'Connor[10,17], Harry C. Dietz[18,19], Joel N. Hirschhorn[4,6,13,20], Heinner Guio[21], Leonid Lecca[14], Eimear E. Kenny[7,8,9], Esther E. Freeman[22], Megan B. Murray[16] & Soumya Raychaudhuri[1,2,3,4,5,23] ✉

Both: changed growth patterns

Open question: is this **genetic** (reflecting long-term adaptation through natural selection) or developmental plasticity?

The problem
Your solutions
Natural selection's solutions

Adaptive responses to high-altitude hypoxia

Homeostasis + acclimation

	Sojourners	Tibetans	Andeans
Resting ventilation	↑	↑	—
Hb concentration	↑	—	↑
Blood flow	↑	↑ ↑	↓
Capillary density	↑	↑ ↑	↑
Stature		↓	↓
Lung volume		↑	↑

Birth weight Offspring survival	↓	—	—

Key

Tibetans:

Dampened erythrocytosis acclimation response

Strategy based on blood flow

Andeans:

Strategy based on erythrocytosis

Both: changed growth patterns

Which of these responses is underpinned by genetics?

Tibetans:

Dampened erythrocytosis acclimation response

Strategy based on blood flow

Andeans:

Strategy based on erythrocytosis

Which of these responses is underpinned by genetics?

Tibetans:

Dampened erythrocytosis acclimation response

Strategy based on blood flow

Andeans:

Strategy based on erythrocytosis

Hypoxia-inducible factors

EGLN1 → PHD2

Also selection on/Around here

Tibetans:

Dampened erythrocytosis acclimation response

Natural selection on *EPAS1* (*HIF2α*) associated with low hemoglobin concentration in Tibetan highlanders

Cynthia M. Beall[a,1], Gianpiero L. Cavalleri[b,1], Libin Deng[c,2], Robert C. Elston[d], Yang Gao[c], Jo Knight[e,f], Chaohua Li[c], Jiang Chuan Li[g], Yu Liang[h], Mark McCormack[b], Hugh E. Montgomery[i,1], Hao Pan[c], Peter A. Robbins[j,1,3], Kevin V. Shianna[k], Siu Cheung Tam[l], Ngodrop Tsering[m], Krishna R. Veeramah[n], Wei Wang[h], Puchung Wangdui[m], Michael E. Weale[e,1], Yaomin Xu[o], Zhe Xu[c], Ling Yang[h], M. Justin Zaman[p], Changqing Zeng[c,1,3], Li Zhang[o,1], Xianglong Zhang[c], Pingcuo Zhaxi[h,1,4], and Yang Tang Zheng[q]

Signals of natural selection for Tibetans → low Hb concentration

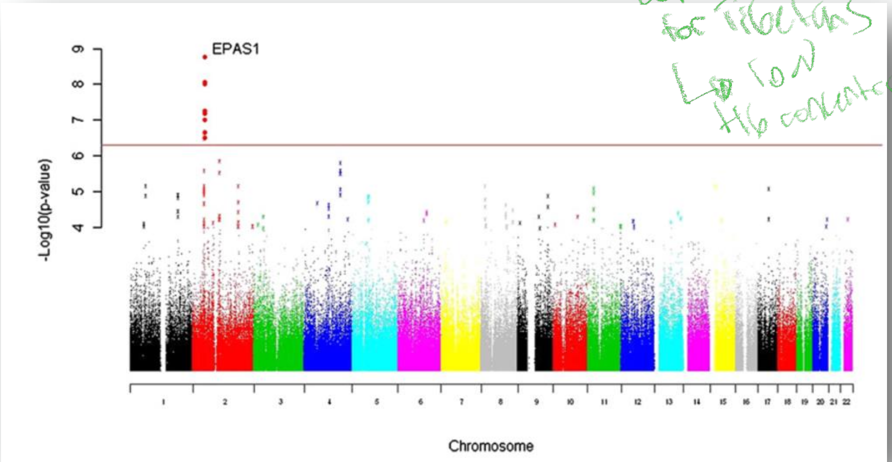
EPAS1 → HIF2A

HIF3A

Transcription factors → *Target genes/proteins*

Erythropoietin (erythrocytosis)

Transferin (transfer of iron, necessary for haemoglobin)



Hypoxia-inducible factors

Ducks also have these adaptations

EGLN1

PHD2

dent on

HIF1A

HIF2A

HIF3A

Transcription factors

Tibetans:

Dampened erythrocytosis acclimation response

Erythropoietin
(erythrocytosis)

Transferin
(transfer of iron, necessary for haemoglobin)

Target genes/proteins

Graham & McCracken *Heredity* (2019)

Hypoxia-inducible factors

EGLN1

PHD2

Degradation, dependent on oxygen availability

HIF1A

EPAS1 → HIF2A

HIF3A

Transcription factors

Tibetans:

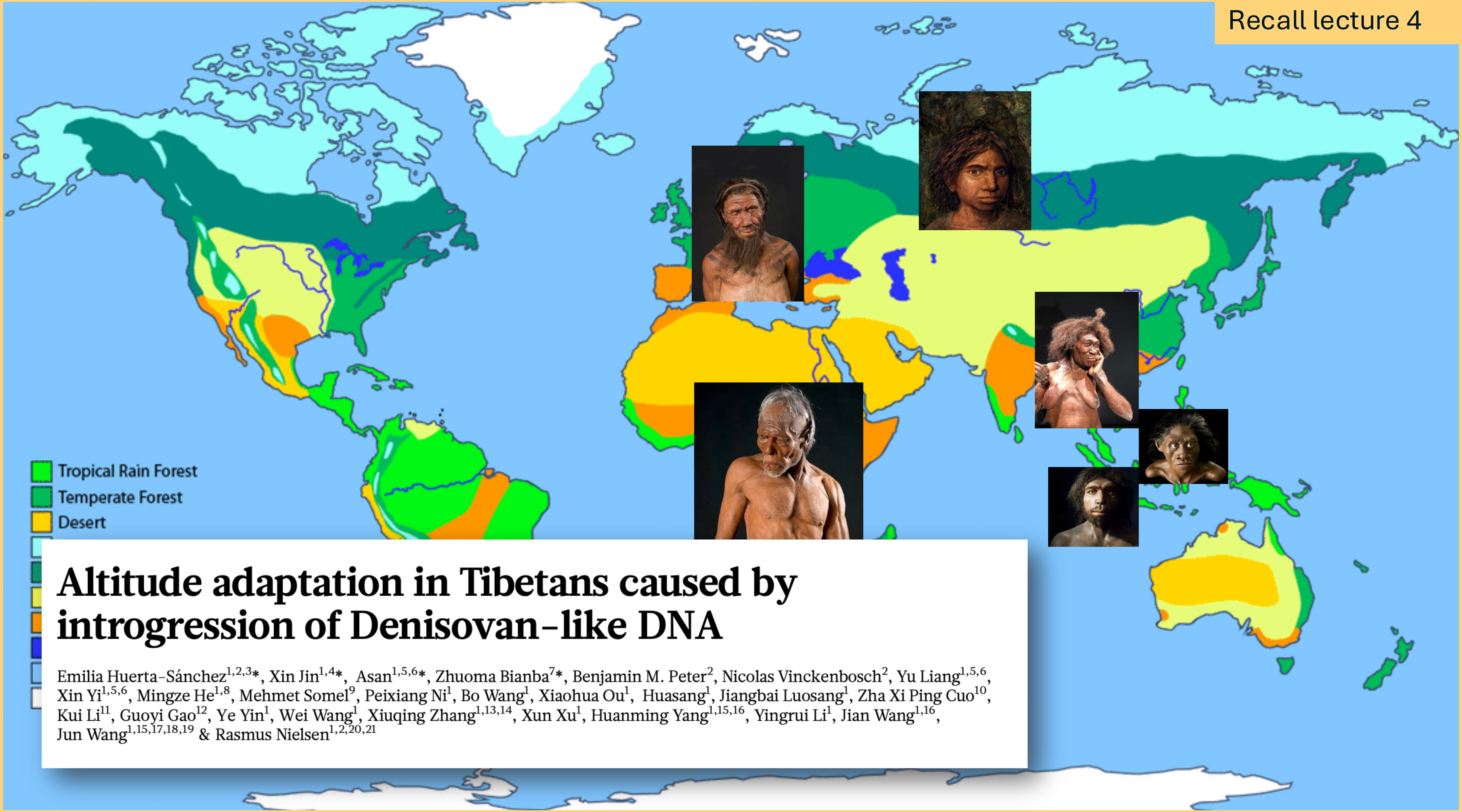
Dampened erythrocytosis acclimation response

Erythropoietin
(erythrocytosis)

Transferin
(transfer of iron, necessary for haemoglobin)

Target genes/proteins

Recall lecture 4



Altitude adaptation in Tibetans caused by introgression of Denisovan-like DNA

Emilia Huerta-Sánchez[1,2,3*], Xin Jin[1,4*], Asan[1,5,6*], Zhuoma Bianba[7*], Benjamin M. Peter[2], Nicolas Vinckenbosch[2], Yu Liang[1,5,6], Xin Yi[1,5,6], Mingze He[1,8], Mehmet Somel[9], Peixiang Ni[1], Bo Wang[1], Xiaohua Ou[1], Huasang[1], Jiangbai Luosang[1], Zha Xi Ping Cuo[10], Kui Li[11], Guoyi Gao[12], Ye Yin[1], Wei Wang[1], Xiuqing Zhang[1,13,14], Xun Xu[1], Huanming Yang[1,15,16], Yingrui Li[1], Jian Wang[1,16], Jun Wang[1,15,17,18,19] & Rasmus Nielsen[1,2,20,21]

The problem
Your solutions
Natural selection's solutions

Key

Drawing on knowledge from past lectures…

28 Aug	Introduction and key frameworks
2 Sep	Adaptation, evolution, and diversity I
4 Sep	Adaptation, evolution, and diversity II
5 Sep	
9 Sep	The deep roots of human biology I
11 Sep	The deep roots of human biology II
12 Sep	
16 Sep	Studying our species: history and ethics
18 Sep	Genetic diversity in modern humans
19 Sep	
23 Sep	Local adaptation
25 Sep	Culture and genetic diversity
26 Sep	
30 Sep	Genes=/=phenotypes
2 Oct	Body shape and size: adaptations to temperature stress?
3 Oct	
7 Oct	Skin and hair: adaptations to solar radiation?

Adaptation framework:

- Hypoxia an extreme stressor
- Homeostatic & acclimating responses
- Growth: Developmental plasticity vs genetics

Deeply conserved pathways
Out-of-Africa & meeting other species

Great example of local adaptation
Alongside disease, hypoxia the strongest selective factor

Growth: developmental plasticity vs genetics

Key

Drawing on knowledge from past lectures…**and building on them**

28 Aug	Introduction and key frameworks
2 Sep	Adaptation, evolution, and diversity I
4 Sep	Adaptation, evolution, and diversity II
5 Sep	
9 Sep	The deep roots of human biology I
11 Sep	The deep roots of human biology II
12 Sep	
16 Sep	Studying our species: history and ethics
18 Sep	Genetic diversity in modern humans
19 Sep	
23 Sep	Local adaptation
25 Sep	Culture and genetic diversity
26 Sep	
30 Sep	Genes=/=phenotypes
2 Oct	Body shape and size: adaptations to temperature stress?
3 Oct	
7 Oct	Skin and hair: adaptations to solar radiation?

Adaptation framework:

- Hypoxia an extreme stressor
- Homeostatic & acclimating responses → Maladaptive
- Growth: Developmental plasticity vs genetics

Deeply conserved pathways

Out-of-Africa & meeting other species

↳ Denisovan admix in Tibetans

Great example of local adaptation → Divergence

Alongside disease, hypoxia the strongest selective factor

↑ Due to amount of time to evolve adaptations

Growth: developmental plasticity vs genetics

The problem

Your solutions

Natural selection's solutions